Statistics for High Dimensional Data Project

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Abstract

This study investigates the spatio-temporal dynamics of NO_2 and CO concentrations in the Beijing area using D-STEM software [5]. After Exploratory Data Analysis and feature engineering, the Hidden Dynamic Geostatistical Model (HDGM) was used to capture spatial and temporal dependencies. Model validation was performed using a Leave-One-Gauge-Out Cross-Validation (LOGOCV) procedure proposed in [2], where each monitoring station was iteratively removed from the training set and its observations predicted. Performance was assessed using standard evaluation metrics and residual analysis.

1 Introduction

Nitrogen dioxide (NO_2) is a gas primarily produced by high-temperature combustion processes, such as those associated with heating systems, vehicle traffic, and industrial activities. Due to its higher density than air, it tends to concentrate in the lower layers of the atmosphere.
Carbon monoxide (CO), on the other hand, is a pollutant generated by incomplete combustion. Its sources can be both natural, such as wildfires and volcanic activity, and anthropogenic, including traffic, domestic heating, and industrial operations.
Both pollutants are highly harmful to human health. For this reason, they are continuously monitored and regulated by specific guidelines established by the World Health Organization (WHO) regarding allowable atmospheric concentrations.
In this context, meteorological variables play a crucial role, as they influence the transport, dispersion, and deposition mechanisms of atmospheric pollutants [1].
This study aims to model the spatio-temporal dynamics of NO_2 and CO concentrations in the Beijing area using the D-STEM framework. In particular, a Hidden Dynamic Geostatistical Model (HDGM) is employed to account for spatial and temporal dependencies.

2 Data

2.1 Description

The dataset was obtained from Kaggle [3] and contains hourly air pollution measurements collected from 12 national monitoring stations in Beijing, together with meteorological data provided by nearby weather stations of the China Meteorological Administration. The observations cover the period from March 1, 2013, to February 28, 2017. The variables included in the dataset are described as follows:

- **No**: Row index.
- **year, month, day, hour**: Temporal components of the observation.
- **$PM_{2.5}$, PM_{10}**: Concentration of fine and coarse particulate matter ($\mu g/m^3$).
- **SO_2, NO_2, O_3, CO**: Concentrations of sulfur dioxide, nitrogen dioxide, ozone and carbon monoxide($\mu g/m^3$).
- **TEMP**: Ambient air temperature (°C).
- **PRES**: Atmospheric pressure (hPa).
- **DEWP**: Dew point temperature (°C).
- **RAIN**: Hourly precipitation (mm).

- **wd**: wind direction (categorical).
- **WSPM**: Wind speed (m/s).
- **station**: Name of the monitoring station providing the data.

2.2 Data Preprocessing

First, a unified timestamp variable *Date* was created for each station by integrating the hour, day, month, and year columns. The observations were then chronologically ordered to ensure temporal consistency.
A temporal subset of the dataset was selected, covering the period from 23/04/2015 to 28/02/2017. This selection ensured complete data availability across all monitoring stations. The resulting dataset consists of 12 stations and 195,180 observations.
A small number of missing values were present in the pollutant variables. In particular, missing values for NO_2 and CO ranged between approximately 0.9% and 3.7% across stations. Since the proportion of missing data was limited, no additional imputation was applied to these variables.
Missing values in meteorological variables (TEMP, PRES, DEWP, and RAIN) were handled using the MATLAB `fillmissing` function with linear interpolation, as the missing observations were sparse and limited in number.
To incorporate spatial information into the D-STEM models, geographic coordinates (latitude, longitude, and altitude) were added for each monitoring station. Latitude and longitude were retrieved from a previous study [4], while altitude values were obtained using Google Earth based on the corresponding geographic coordinates.
The resulting geographic information is reported in Table 1.

Table 1: Geographic information of monitoring stations

Station	**Longitude**	**Latitude**	**Altitude (m)**
Aotizhongxin	116.397	39.982	47
Changping	116.230	40.217	67
Dingling	116.220	40.292	126
Dongsi	116.417	39.929	49
Guanyuan	116.339	39.929	51
Gucheng	116.184	39.914	76
Huairou	116.628	40.328	59
Nongzhanguan	116.461	39.937	40
Shunyi	116.655	40.127	41
Tiantan	116.407	39.886	45
Wanliu	116.287	39.987	47
Wanshouxigong	116.352	39.878	47

2.3 Exploratory Data Analysis

The spatial distribution of the monitoring stations within the Beijing area was initially analyzed. The mapping reveals a heterogeneous layout: eight stations are concentrated in the high-density urban center, while the remaining four are located in suburban and peripheral areas. This classification is consistent with the grouping adopted in [6].

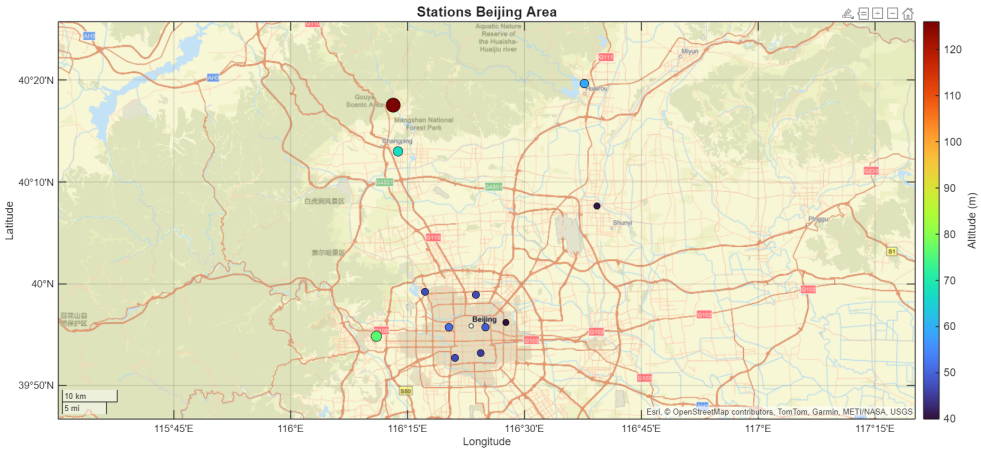

Figure 1: Beijing stations map

A comparative analysis was then performed to examine the characteristics of NO2 and CO concentrations across the two identified clusters. This comparison aims to highlight differences in pollutant dynamics and assess how territorial characteristics influence observed concentrations.
The results for NO2 show a clear contrast between urban and suburban stations, with consistently higher concentrations in urban areas. A pronounced seasonal cycle is also visible, with higher values in winter and lower values in summer. This pattern is likely due to increased heating emissions and reduced atmospheric dispersion during colder months.

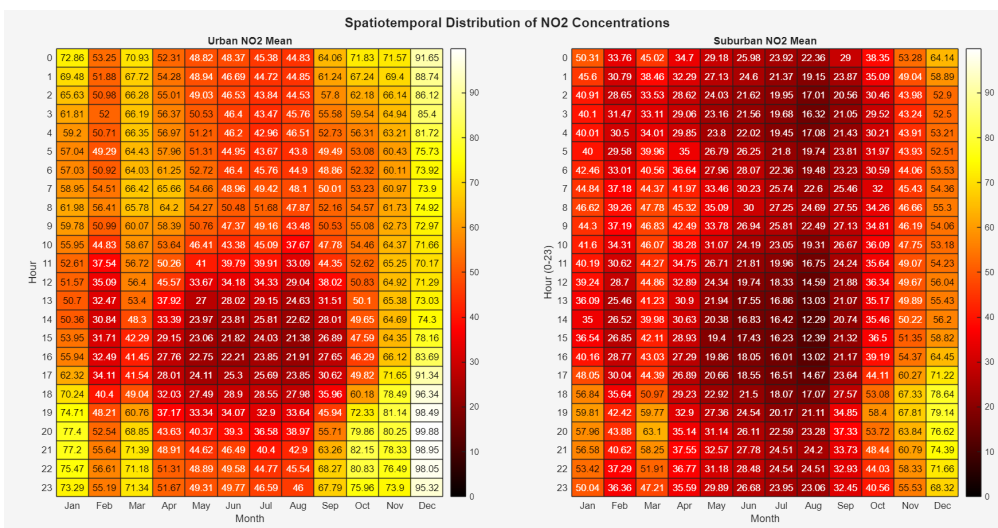

Figure 2: Mean NO2 concentrations by month and hour for urban and suburban stations

To further investigate seasonal behaviour, mean pollutant concentrations were aggregated by season, distinguishing again between urban and suburban areas. The results confirm a strong seasonal pattern and consistently higher levels in urban locations.

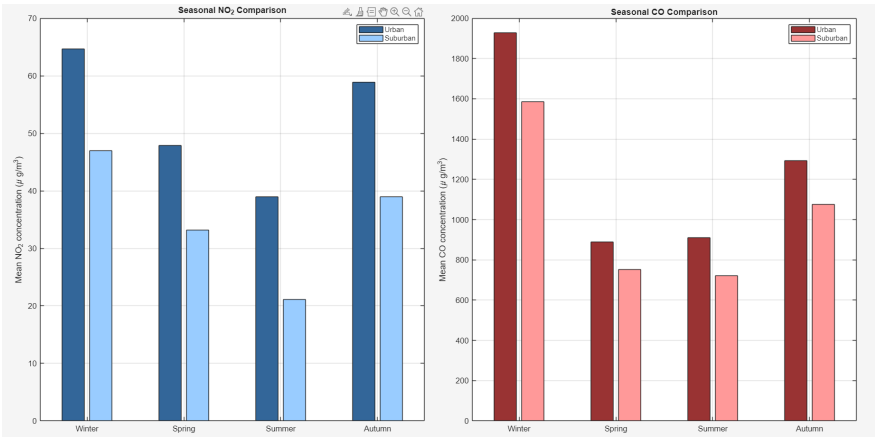

Figure 3: Seasonal comparison of pollutant concentrations between urban and suburban stations

Finally, Pearson correlation analysis was performed on numerical variables. The results indicate a strong correlation among key meteorological variables, particularly *DEWP*, *TEMP*, and *PRES*. This suggests potential multicollinearity, which should be considered when including these covariates in the model.

Figure 4: Correlation Heatmap

2.4 Feature Engineering

Based on the EDA results, two variables, *sea_sin* and *sea_cos*, were created to capture seasonal effects. Seasonality was modeled using sine and cosine functions to ensure continuity across months. These variables were defined as:

$$sea_sin = \sin\left(\frac{2\pi(month-1)}{12}\right), \qquad sea_cos = \cos\left(\frac{2\pi(month-1)}{12}\right)$$

To better represent precipitation effects, a new variable *RAIN_24h* was introduced as the cumulative rainfall over the previous 24 hours:

$$RAIN_24h = \text{movsum}(RAIN, [23\ 0])$$

To address high correlation among meteorological covariates(*TEMP*, *DEWP*, *PRES*)and to mitigate multicollinearity, *relative_humidity* was derived. By combining temperature and dew point effects into a single indicator, the total number of predictors is reduced while retaining essential physical information. *relative_humidity* was calculated using the *Magnus–Tetens* approximation:

$$RH = 100 \cdot \exp\left(\frac{aT_d}{b+T_d} - \frac{aT}{b+T}\right)$$

where T is the air temperature, T_d is the dew point temperature, $a = 17.625$, and $b = 243.04$.
Additional binary indicators were introduced to capture specific temporal patterns:

- *is_working_hour*: identifies observations during standard working hours (09:00–18:00).
- *is_weekend*: identifies observations collected on Saturdays and Sundays.

2.5 Feature Selection

Feature selection was performed based on correlation analysis to identify highly linearly correlated predictors. In addition, the Variance Inflation Factor (VIF) was computed to assess potential multicollinearity issues among the covariates.
The combination of correlation analysis and VIF values allowed the identification and removal of redundant variables. In particular, *TEMP* and *DEWP* were removed due to their high correlation and replaced with the derived variable *relative_humidity*.
Furthermore, *PRES* was excluded because it showed strong correlation with *sea_cos*.
Similarly, *RAIN* was replaced by the cumulative precipitation variable *RAIN_24h*, which better captures temporal accumulation effects.
The final set of covariates is:

relative_humidity, *WSPM*, *RAIN_24h*, *sea_sin*, *sea_cos*, *latitude*, *longitude*, *altitude*, *is_weekend*, *is_working_hour*

This set provides a balanced representation of spatial, temporal, and meteorological covariates.

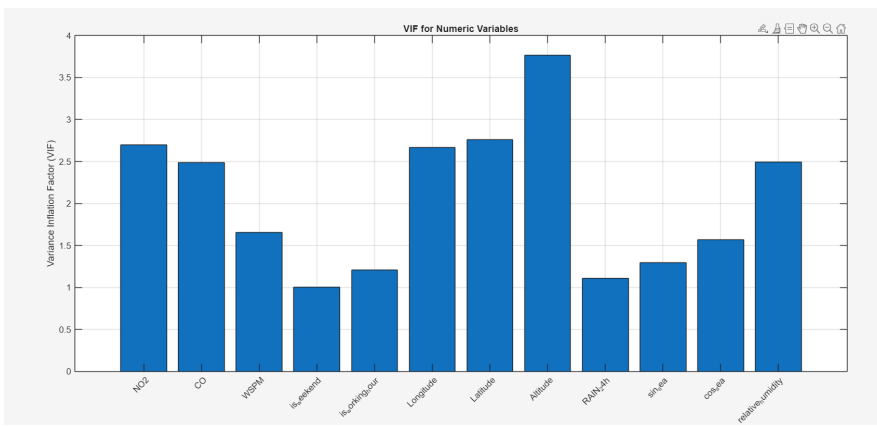

Figure 5: VIF values after feature selection

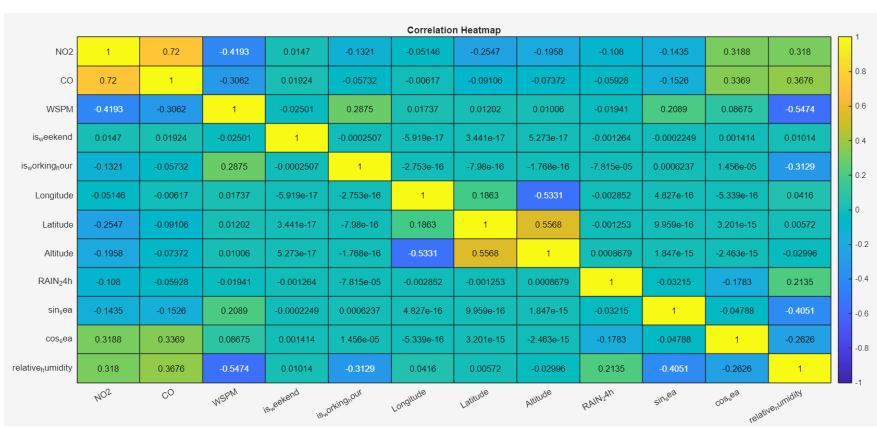

Figure 6: Correlation heatmap after feature selection

3 Methodology

The main objective of this study is to evaluate and compare different modeling approaches for air pollution prediction by considering alternative sets of covariates. In particular, models of increasing complexity are developed to assess the contribution of both model structure and predictor selection.
All models are validated using a Leave-One-Gauge-Out Cross-Validation (LOGOCV) procedure. This choice is motivated by the limited number of monitoring stations (12 stations) and by the results of the Exploratory Data Analysis, which highlighted a strong influence of the spatial zone on temporal patterns. Therefore, LOGOCV provides a more reliable assessment of the models spatial generalization ability.
Model performance is evaluated using four metrics: the mean Root Mean Squared Error (RMSE), the mean coefficient of determination (R^2), the mean autocorrelation of residuals at lag 1 (ACF(1)), and the mean coefficient of variation of the RMSE (CVRMSE).
All metrics are computed across the Leave-One-Gauge-Out Cross-Validation (LOGOCV) folds and then averaged to obtain a single overall performance measure.
The CVRMSE is defined as the mean RMSE divided by the mean value of the pollutant in the dataset.
The ACF(1) metric measures the first-lag autocorrelation of model residuals and is used to assess potential temporal dependence not captured by the model.
The initial baseline models are constructed using the set of 10 covariates selected during the feature selection stage.
For the univariate HDGM models, two different covariate configurations are considered. In Model M2, only spatial covariates (latitude, longitude, and altitude) are included in order to isolate the contribution of spatial variability. In Model M3, the full set of 10 covariates is used, including spatial, temporal, and meteorological variables.
Finally, in the bivariate HDGM framework (M4), reduced covariate sets are adopted based on the estimated covariates coefficients obtained in Model M3, in order to simplify the model while retaining the most relevant predictors.

Table 2: Overview of the proposed models for NO_2 and CO analysis

Model	Description
M1	Linear regression baseline model
M2	Univariate HDGM with spatial covariates only
M3	Univariate HDGM with spatial, temporal, and meteorological covariates
M4	Bivariate HDGM with reduced covariate sets based on M3 coefficients

3.1 Linear Model

A simple linear regression model is used as a baseline:

$$y(s) = x(s)'\beta + \varepsilon(s) \tag{1}$$

where $s \in R^2$ denotes a generic spatial location, $x(s)$ is the vector of spatial covariates, β is a vector of parameters, and $\varepsilon(s) \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is the random error term.

3.2 Univariate HDGM

The **Hidden Dynamic Geostatistical Model (HDGM)** presented in [2], is defined by the following formulation:

$$y(s,t) = X(s,t)\beta + az(s,t) + \epsilon(s,t) \tag{1}$$
$$z(s,t) = gz(s,t-1) + \eta(s,t) \tag{2}$$

where $z(s,t)$ is a scalar latent process that follows a Markov dynamics governed by the transition coefficient g. The measurement error is $\epsilon(s,t) \sim \mathcal{N}(0, \sigma_\epsilon^2)$, while the innovation term $\eta(s,t)$ is a Gaussian random process with covariance structure:

$$\eta(s,t) \sim \mathcal{GP}(0, \rho(||s - s'||; \theta))$$

The parameter set is given by:

$$\Psi = \{\beta, a, \sigma_\epsilon^2, \theta, g\}$$

The model is estimated via the EM algorithm. In the E-step, the latent process $z(s,t)$ is estimated using a Kalman smoother, which also enables spatial prediction by treating missing observations in the response vector y as unobserved entries.

3.3 Multivariate HDGM

The multivariate extension of the HDGM allows the joint modeling of p correlated pollutants. The model is defined as:

$$y(s,t) = X(s,t)\beta + z(s,t) + \epsilon(s,t) \tag{3}$$
$$z(s,t) = Gz(s,t-1) + \eta(s,t) \tag{4}$$

where $y(s,t)$ and $z(s,t)$ are $p \times 1$ vectors representing the observed variables and the latent spatio-temporal process, respectively. The latent process follows a Markovian dynamic governed by the diagonal transition matrix G of size $p \times p$.
The innovation term is defined as a multivariate Gaussian random process:

$$\eta(s,t) \sim \mathcal{GP}_p(0, V\rho(||s - s'||; \theta))$$

where V is a variance–covariance matrix. The measurement error is:

$$\epsilon(s,t) \sim \mathcal{N}_p(0, \Sigma_\epsilon)$$

The parameter set of the multivariate model is given by:

$$\Psi = \{\beta, \Sigma_\epsilon, \theta, V, G\}.$$

4 Results Analysis

4.1 Linear Models M1

The performance of these models, reported in Tables 3 and 4, show relatively low R^2 values together with high RMSE and CVRMSE values. These results are expected given the simplicity of the model, which is unable to adequately capture the spatio-temporal structure of the data.
Moreover, the very high residual autocorrelation values indicate a strong unmodeled temporal dependence, suggesting that the linear model fails to properly describe the temporal dynamics of the pollutants. These findings motivate the adoption of more advanced spatio-temporal models, such as the HDGM, which explicitly account for spatial dependence and temporal dynamics.

Table 3: Performance M1 NO_2

Metrics	Value
mean RMSE	26.46
mean CVRMSE	0.563
mean R2	0.323
mean ACF 1	0.858

Table 4: Performance M1 CO

Metrics	Value
mean RMSE	964.19
mean CVRMSE	0.80
mean R2	0.393
mean ACF 1	0.894

4.2 Univariate HDGM M2-M3

As shown in Tables 5–8, the HDGM models perform better than the linear baseline model M1, with a clear reduction in RMSE and higher R^2 values for both pollutants. These results are consistent with the HDGM structure, which directly models spatio-temporal dependence. Adding temporal and meteorological covariates in Model M3 leads to a small additional improvement in performance. However, the residual autocorrelation is still relatively high (ACF lag-1 $\approx$ 0.72–0.79), which means that part of the temporal dependence is not fully captured by the model.
The limited improvement from M2 to M3 suggests that spatial structure plays a dominant role in explaining pollutant variability, while additional covariates provide only marginal gains.

Table 5: Performance M2 NO_2

Metrics	Value
mean RMSE	17.79
mean CVRMSE	0.378
mean R2	0.632
mean ACF 1	0.79

Table 6: Performance M2 CO

Metrics	Value
mean RMSE	529.65
mean CVRMSE	0.439
mean R2	0.79
mean ACF 1	0.74

Table 7: Performance M3 NO_2

Metrics	Value
mean RMSE	17.34
mean CVRMSE	0.368
mean R2	0.654
mean ACF 1	0.784

Table 8: Performance M3 CO

Metrics	Value
mean RMSE	510.15
mean CVRMSE	0.423
mean R2	0.81
mean ACF 1	0.725

Table 9: M3 Model: Mean Beta and t-statistics for NO_2 and CO

	NO_2		**CO**	
Covariate	**beta**	**t-stat**	**beta**	**t-stat**
relative_humidity	0.3697	71.0043	0.3592	75.9643
RAIN_24h	-0.1193	14.7369	-0.0498	6.9288
WSPM	-0.0125	6.9260	-0.0050	2.8223
sin_sea	0.0467	2.4205	-0.0229	0.9342
cos_sea	0.3081	15.8526	0.3293	13.2432
Latitude	-0.1293	7.5991	-0.0691	3.5208
Longitude	-0.2238	14.7964	-0.0736	4.2908
Altitude	-0.3213	24.5816	-0.1021	7.2826
is_weekend	0.0125	0.9382	0.0086	0.6918
is_working_hour	-0.0006	0.3313	0.0440	8.9560

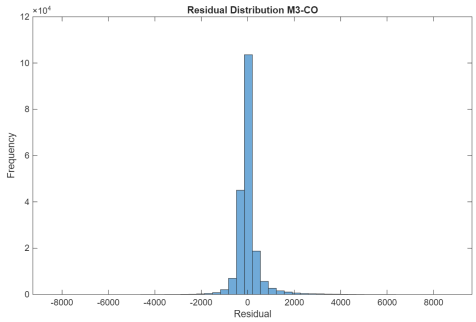

Figure 7: M3-CO Residual Distribution

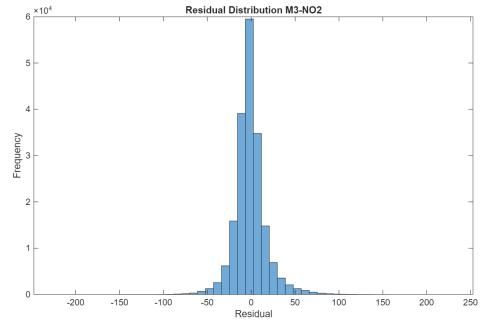

Figure 8: M3-NO_2 Residual Distribution

4.3 Bivariate HDGM (M4)

The final model considered is the bivariate HDGM, where NO_2 and CO are modeled jointly. To ensure model consistency and interpretability, a common set of covariates is used for both pollutants. A reduced set of covariates is used in order to simplify the model while maintaining its predictive performance. The selected covariates are: relative humidity, RAIN_24h, cos_sea, latitude, longitude, and altitude. This choice is based on the estimated coefficients and their statistical significance (t-statistics) obtained in Model M3. Overall, the bivariate HDGM shows stable and competitive performance. Compared to the univariate models, it achieves similar or slightly improved accuracy, particularly for CO, while maintaining a good balance between model complexity and interpretability. However, the residual autocorrelation remains non-negligible, indicating that some temporal dependence is still not fully captured.

Table 10: Performance M4 NO_2

Metrics	**Value**
mean RMSE	17.65
mean CVRMSE	0.375
mean R2	0.64
mean ACF 1	0.76

Table 11: Performance M4 CO

Metrics	**Value**
mean RMSE	493.43
mean CVRMSE	0.409
mean R2	0.83
mean ACF 1	0.69

5 Conclusion

This study analyzed the spatio-temporal behavior of NO_2 and CO using different models with increasing complexity, from a simple linear regression to univariate and bivariate Hidden Dynamic Geostatistical Models (HDGM). The main goal was to understand how spatial, temporal, and meteorological variables affect model performance.

The results show that the linear model (M1) does not describe the data well. It has low R^2 values and strong autocorrelation in the residuals. In contrast, all HDGM models (M2–M4) clearly improve the results in terms of RMSE, R^2, and overall prediction accuracy. This confirms that spatial and temporal dependencies are important for modeling air pollution.

Among the HDGM models, using only spatial covariates (M2) already explains a large part of the variability. Adding temporal and meteorological variables (M3) leads to only small improvements.

The bivariate model (M4) gives similar or slightly better results while using fewer covariates. However, the residual autocorrelation is still high in all models. This means that some temporal patterns are still not fully captured.

References

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